

## Project Design Phase-II

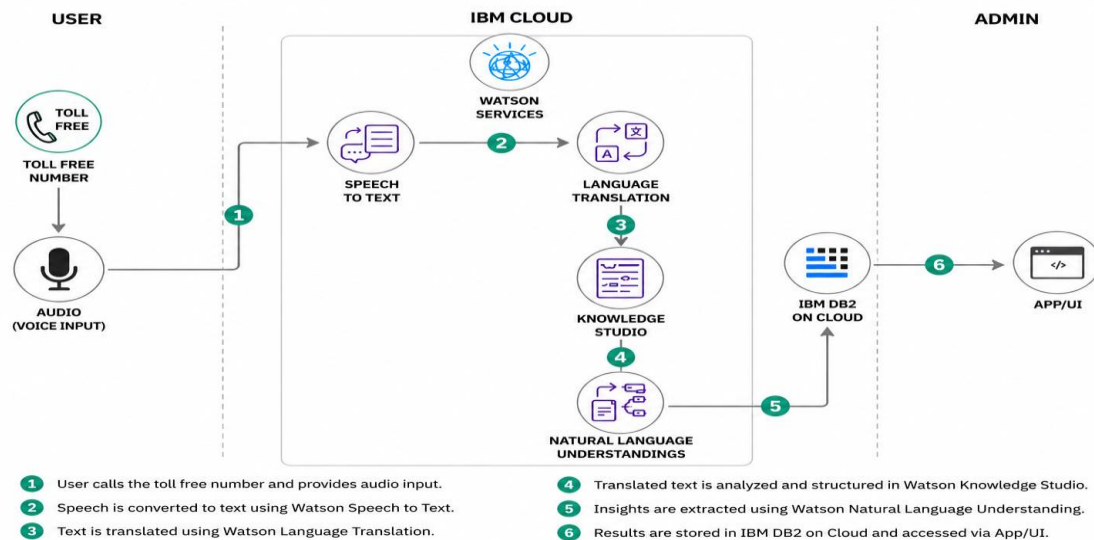
### Technology Stack (Architecture & Stack)

|               |                                               |
|---------------|-----------------------------------------------|
| Date          | 01 July 2026                                  |
| Team ID       |                                               |
| Project Name  | India's Agricultural Crop Production Analysis |
| Maximum Marks | 4 Marks                                       |

#### Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

#### Example: Architecture for India's Agricultural Crop Production Analysis



#### Guidelines:

Include all the processes (As an application logic / Technology Block)

Provide infrastructural demarcation (Local / Cloud)

Indicate external interfaces (third party API's etc.)

Indicate Data Storage components / services

Indicate interface to machine learning models (if applicable)

**Table-1 : Components & Technologies:**

| S.No | Component         | Description                                     | Technology                                          |
|------|-------------------|-------------------------------------------------|-----------------------------------------------------|
| 1.   | User Interface    | How user interacts with application e.g. Web UI | HTML5, CSS3, Bootstrap 5, JavaScript                |
| 2.   | Application Logic | Visualization and interaction processing logic  | Tableau Public Dashboard & Story Engine             |
| 3.   | Database          | Data Type, Configurations etc.                  | Microsoft Excel (.xlsx) Static Data                 |
| 4.   | Infrastructure    | Application Deployment on Cloud                 | Vercel (Web Hosting), Tableau Public (Data Hosting) |

**Table-2: Application Characteristics:**

| S.No | Characteristics          | Description                              | Technology                                         |
|------|--------------------------|------------------------------------------|----------------------------------------------------|
| 1.   | Open-Source Frameworks   | List the open-source frameworks used     | Bootstrap 5 CSS Framework                          |
| 2.   | Security Implementations | Security / access controls implemented   | Vercel Auto SSL/TLS Encryption                     |
| 3.   | Scalable Architecture    | Justify the scalability of architecture  | Tableau Public CDN distributed architecture        |
| 4.   | Performance              | Design consideration for the performance | In-memory Tableau data engine processing 345k rows |

**References:**

<https://www.tableau.com/developer>

<https://data.gov.in/>

<https://agriwelfare.gov.in/>

<https://getbootstrap.com/docs/>

<https://vercel.com/docs>